5 Review

5.1 Voting Theory

Do plurality method.

Example 5.1.1 Let's look the following vote.

	1	3	3	3
1st Choice	A	A	O	H
2nd Choice	O	H	H	A
3rd Choice	H	O	A	O

Example 5.1.2 Perform the Pairwise comparison method with the following preference schedule.

	5	5	6	4
1st	D	A	C	B
2nd	A	C	B	D
3rd	C	B	D	A
4th	B	D	A	C

If D drops out, that affects who wins! This violates the IIA criterion.

Example 5.1.3 Who wins the following election under the plurality with elimination method?

	37	22	12	29
1st	A	B	B	C
2nd	B	C	A	A
3rd	C	A	C	B

C is eliminated first, then A wins with the majority.

Now, suppose that after officials have announced the results, the ballots were accidentally destroyed before they could be certified, so the votes have to be recast. Ten of the people whose preference was B, C, A decide that they want to now support the winner A, so they change their votes to A, B, C, all the other votes remain the same. What are the results of the election now?

1st				
2nd				
3rd				

Discuss the monotonicity criterion, this is a violation of it.

Example 5.1.4 Use Borda Count method to determine the winner of the election.

	51	25	10	14
1st	S	T	P	O
2nd	T	P	T	T
3rd	O	O	O	P
4th	P	S	S	S

Notice that T wins but someone else has a majority, review majority criterion.

5.2 Weighted Voting

Example 5.2.1 Determine all the players who have veto power in the following weighted voting system:

$$[34 : 12, 11, 10, 7, 3].$$

Players 1, 2, and 3 all have veto power.

Definition 5.2.2 To calculate the Banzhaf Power Index:

- List all winning coalitions.
- In each winning coalition, determine all players who are critical.
- Count up how many times each play is critical.
- Convert these to fractions by dividing by the total times any player is critical.

Example 5.2.3 Calculate the Banzhaf Power Index for the three players in the following weighted voting system

$$[10 : 7, 6, 2].$$

Example 5.2.4 Calculate the Banzhaf Power Index for the following weighted voting system

$$[8 : 6, 3, 2].$$

$\boxed{P_1 = 3/5,\ P_2 = 1/5 \text{ and } P_3 = 1/5.}$

Definition 5.2.5 (Shapley-Shubik power index) To compute the shapley-shubik power index:

- List all sequential coalitions
- Determine all pivotal players
- Count up how many times each player was critical
- convert these into fractions by deviding each count by the total number of sequential coalitions

Example 5.2.6 Calculate the Shapley-Shubik Power Index for the following weighted voting system

$$[8 : 6, 3, 2].$$

5.3 Apportionment

lower quotas are rounded down.

Definition 5.3.1 For **Hamilton's Method**, we perform the following steps:

- Calculate the **divisor**: (total population divided by total number of representatives)
- Calculate the **quota** for each state: (state's population divided by divisor)
- Calculate **lower quotas**
- Assign any remaining representatives to those states whose decimal parts of the quota was largest.

Example 5.3.2 The state of Delaware has three counties: Kent, New Castle, and Sussex. The Delaware state House of Representatives has 41 members. If Delaware wants to divide this representation along county lines, let's use Hamilton's method to apportion them. The populations of the counties are as follows (from the 2010 Census): Kent: 162,310, New Castle: 538,479, and Sussex: 197,145.

total pop is 897,934.

Definition 5.3.3 Jefferson's method:

- Find the Standard divisor
- Find each states lower quota
- If the lower quotas do not add up to the number of seats available, then adjust the divisor, and go back to step 2.

Example 5.3.4 Let's redo the first example from last time, this time with Jefferson's method.

County	Population
Kent	162,310
New Castle	538,479
Sussex	197,145
Total	1,052,567

5.4 Fair Division

Example 5.4.1 Suppose that Betty and Cade are sharing a cake which is half strawberry and half vanilla. If there are six slices of cake total, and cade values straberry 3 times as much as he values vanilla, how much is each slice of straberry worth to him?

Example 5.4.2 Troy and Abed are sharing a Subway sandwich, which is a foot long and half veggie and half chicken. Troy likes chicken 2 times as much as veggies, and Abed likes chicken and veggies equally. If Troy is the divider, describe how he will split the sandwich (only vertical slices are allowed). Which part will Abed choose?

Definition 5.4.3 (The Lone Divider Method)

1. The divider divides the item into N pieces, which we'll label s_1, s_2, ..., s_N.
2. Each of the choosers will separately list which pieces they consider to be a fair share. This is called their bid.
3. The lists are examined. There are two possibilities:
   (a) If it is possible to assign each player a piece that they bid on, then do so, and the divider gets the remaining piece.
   (b) If this is not possible, then the divider receives an uncontested piece, and the remaining $N-1$ parties repeat the entire procedure. If there are only 2 players left, they may use the Divider Chooser method.

Example 5.4.4 Consider the following example. Who is the divider? Is there a fair division for this example? Why does the Lone Divider method guarantee a fair division in this example?

	Piece 1	Piece 2	Piece 3
Chad	40%	30 %	30 %
Brody	45%	30%	25%
Mary	33.3%	33.3%	33.3%

Definition 5.4.5 The method begins by compiling a list of items to be divided. Then:

1. Each party involved lists, in secret, a dollar amount they value each item to be worth. This is their sealed bid.
2. The bids are collected. For each party, the value of all the items is totaled, and divided by the number of parties. This defines their fair share.
3. Each item is awarded to the highest bidder.
4. For each party, the value of all items received is totaled. If the value is more than that party's fair share, they pay the difference into a holding pile. If the value is less than that party 's fair share, they receive the difference from the holding pile. This ends the initial allocation.
5. In most cases, there will be a surplus, or leftover money, in the holding pile. The surplus is divided evenly between all the players. This produces the final allocation.

Example 5.4.6 Jamal, Maggie, and Kendra are dividing an estate consisting of a house, a vacation home, and a small business. Their valuations (in thousands) are shown below. Determine the final allocation.

	Jamal	Maggie	Kendra
House	\$250	\$300	\$280
Vacation Home	\$170	\$180	\$200
Small Business	\$300	\$255	\$270